Software Development Life Cycle (SDLC)

What is SDLC?

Definition:

Software Development Life Cycle (SDLC) is a structured process used to plan, design, develop, test, deploy, and maintain software applications in a systematic way.

Simple Example

Imagine building a house:

- First, understand requirements.
- Create a blueprint.
- Construct the house.
- Inspect for defects.
- Hand over to the owner.
- Maintain and repair when needed.

Similarly, software is developed using SDLC.

Why Do We Use SDLC?

Without SDLC:

❌ Requirements get missed
❌ More bugs occur
❌ Project delays happen
❌ Budget increases
❌ Customer dissatisfaction

With SDLC:

✅ Clear planning
✅ Better quality software
✅ Reduced risks
✅ Timely delivery
✅ Easier maintenance

Advantages of SDLC

1. Better Project Planning

Helps estimate cost, effort, and timeline.

Example:
Banking application development planned for 8 months.

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2. Improved Quality

Testing is performed at every stage.

Example:
Detecting login page defects before production.

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3. Reduced Risk

Risks are identified early.

Example:
Security issues found during design phase.

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4. Cost Control

Fixing issues early is cheaper.

Example:
Changing a requirement during design costs less than after deployment.

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5. Better Communication

All teams work with clear documentation.

Example:
Business Analyst shares requirements with developers.

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6. Faster Delivery

Work is organized phase-wise.

Example:
Teams complete modules according to schedule.

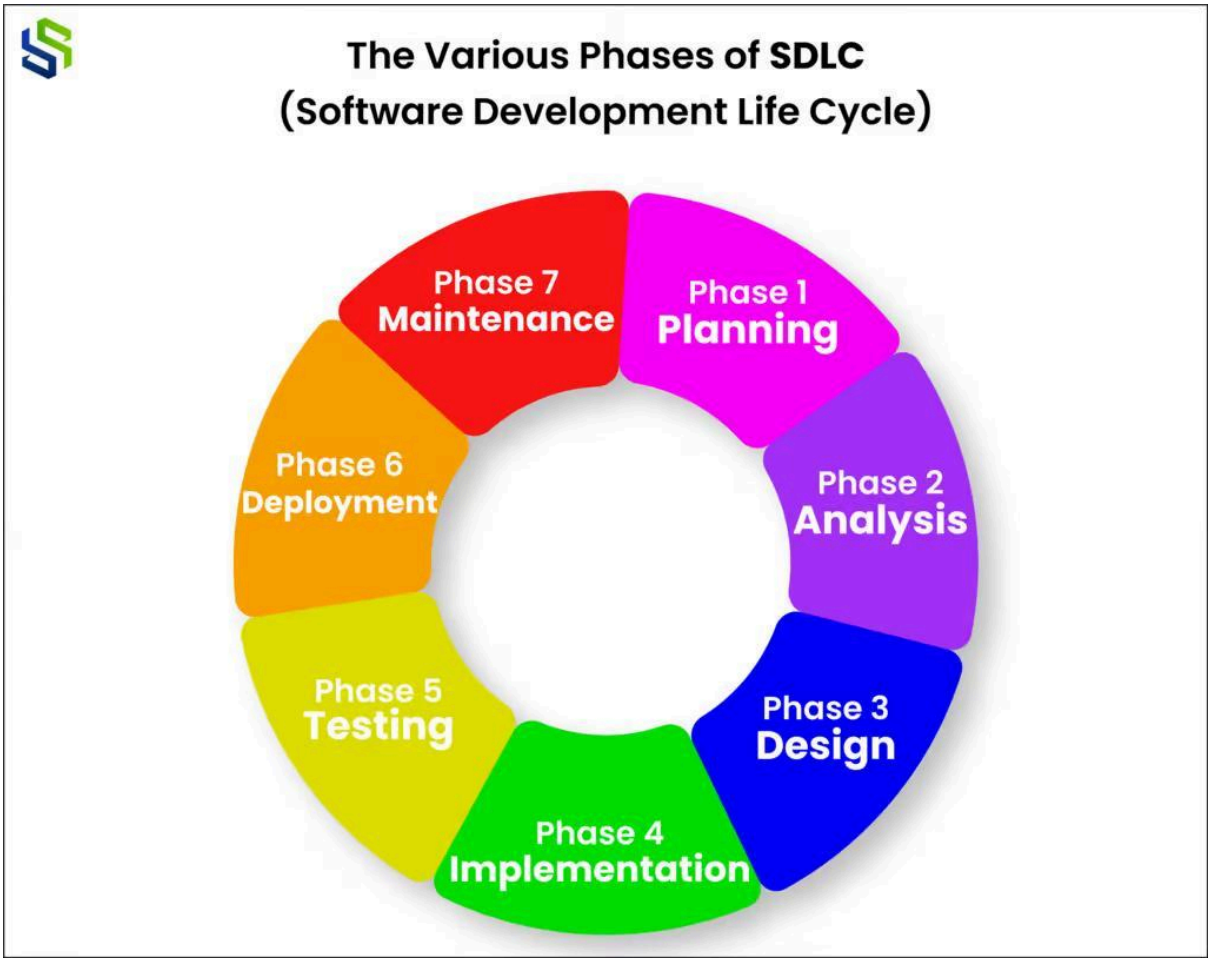
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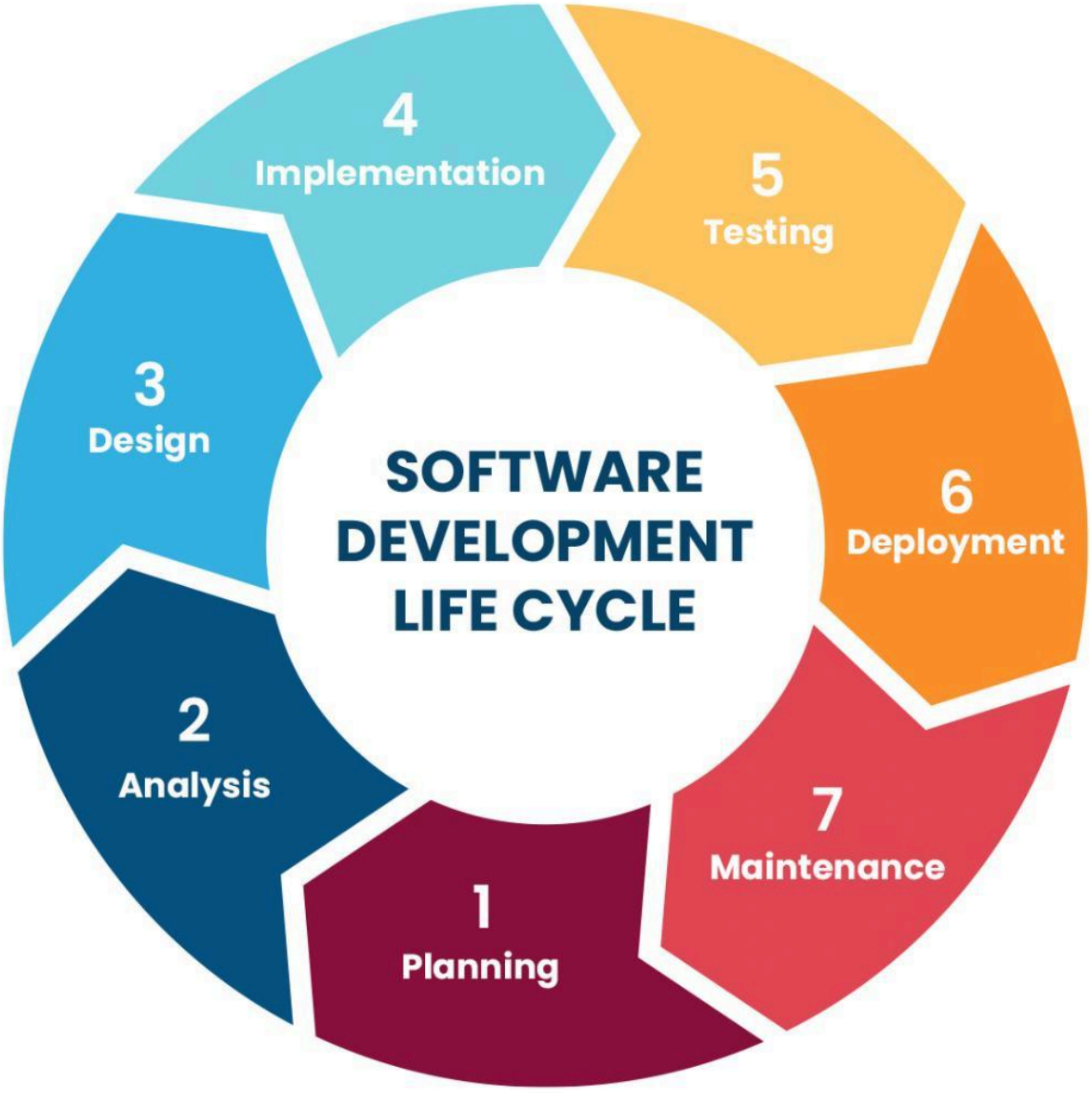
7. Easier Maintenance

Changes can be managed effectively.

Example:
Adding UPI payment feature after deployment.

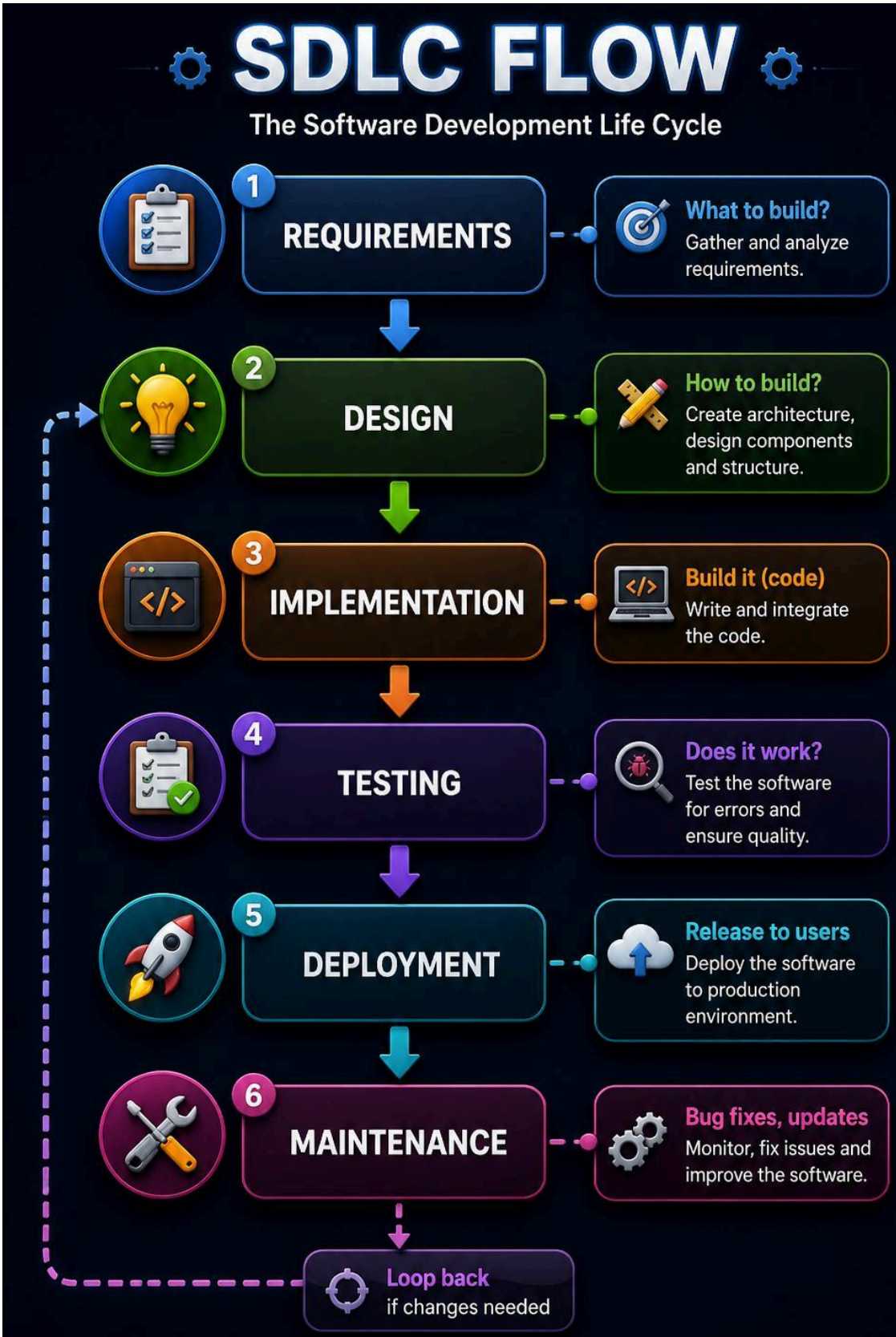
SDLC Lifecycle




SOFTWARE DEVELOPMENT LIFE CYCLE
1 Planning
2 Analysis
3 Design
4 Implementation
5 Testing
6 Deployment
7 Maintenance

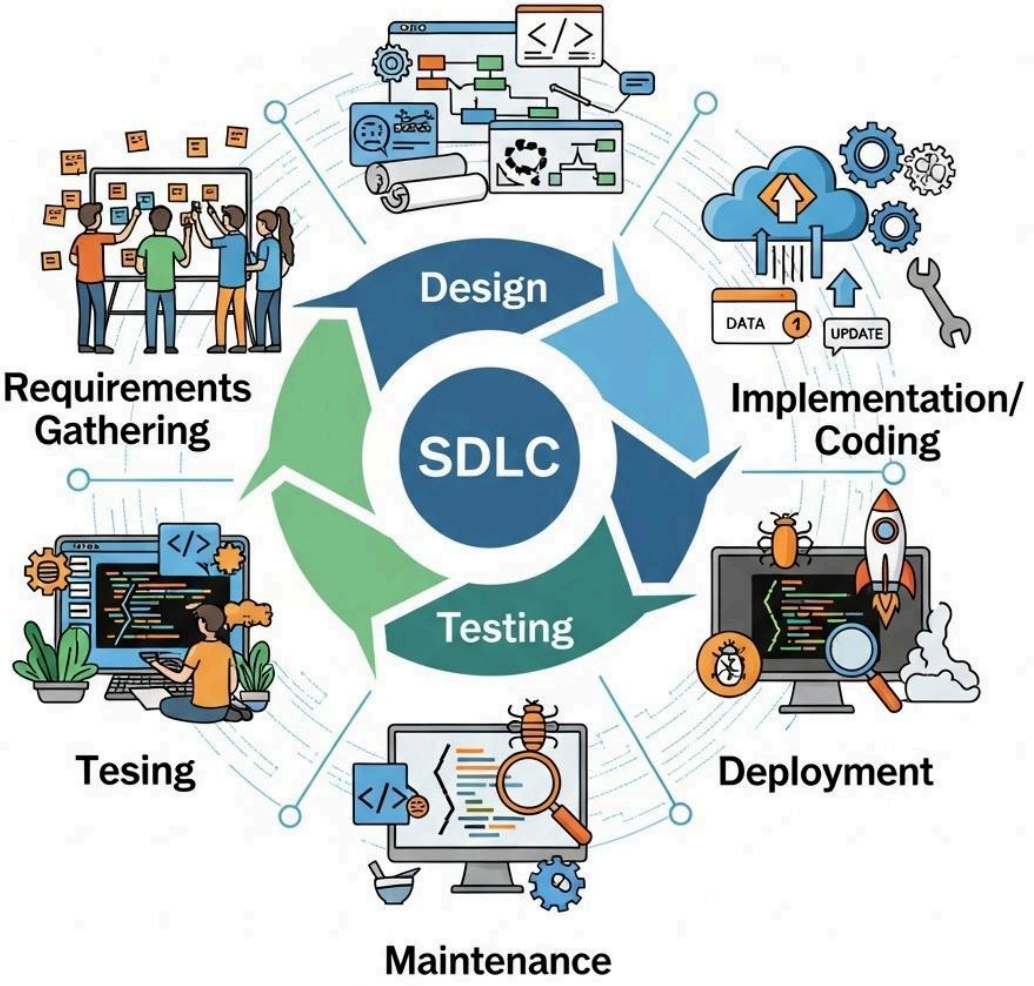
SDLC FLOW

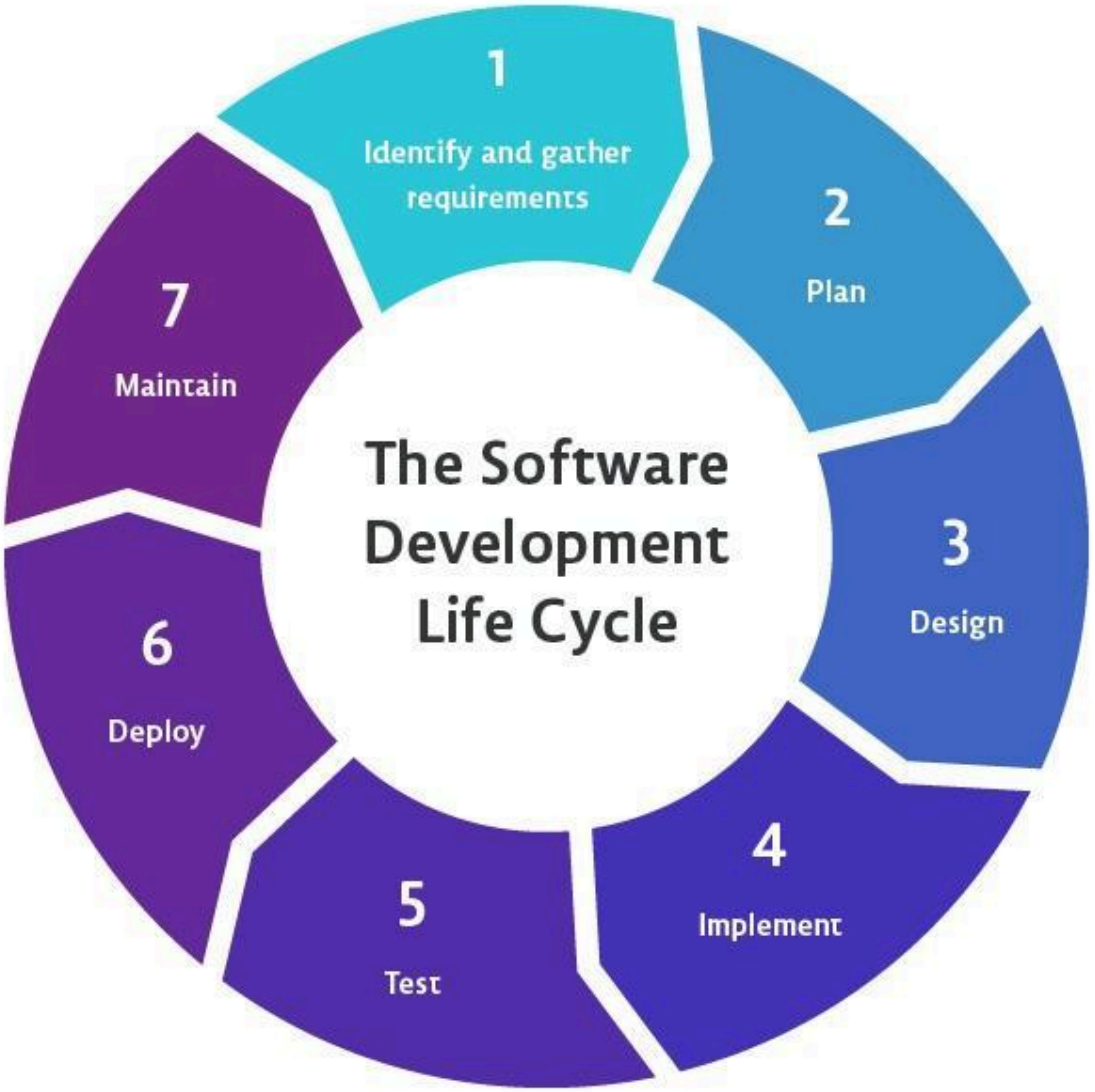
The Software Development Life Cycle

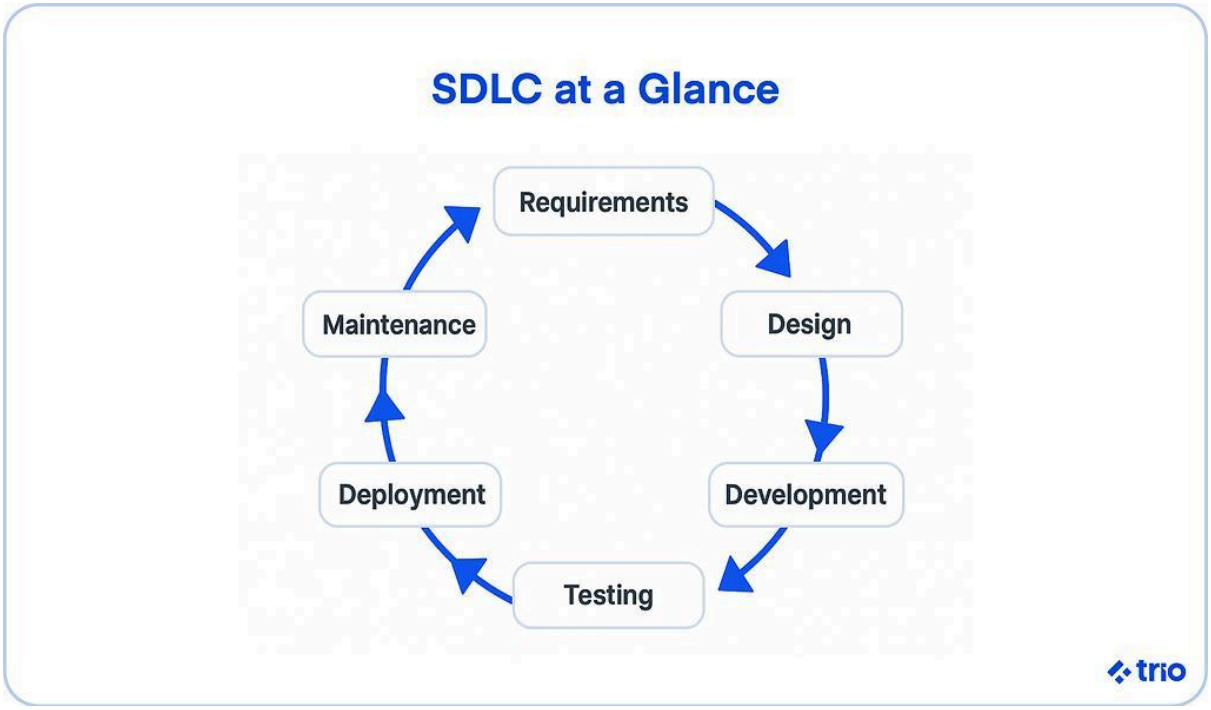


SDLC is an iterative process that helps in building high-quality software efficiently.

Software Development Lifecycle




1
Identify and gather requirements
2
Plan
3
Design
4
Implement
5
Test
6
Deploy
7
Maintain
The Software Development Life Cycle



SDLC Flow

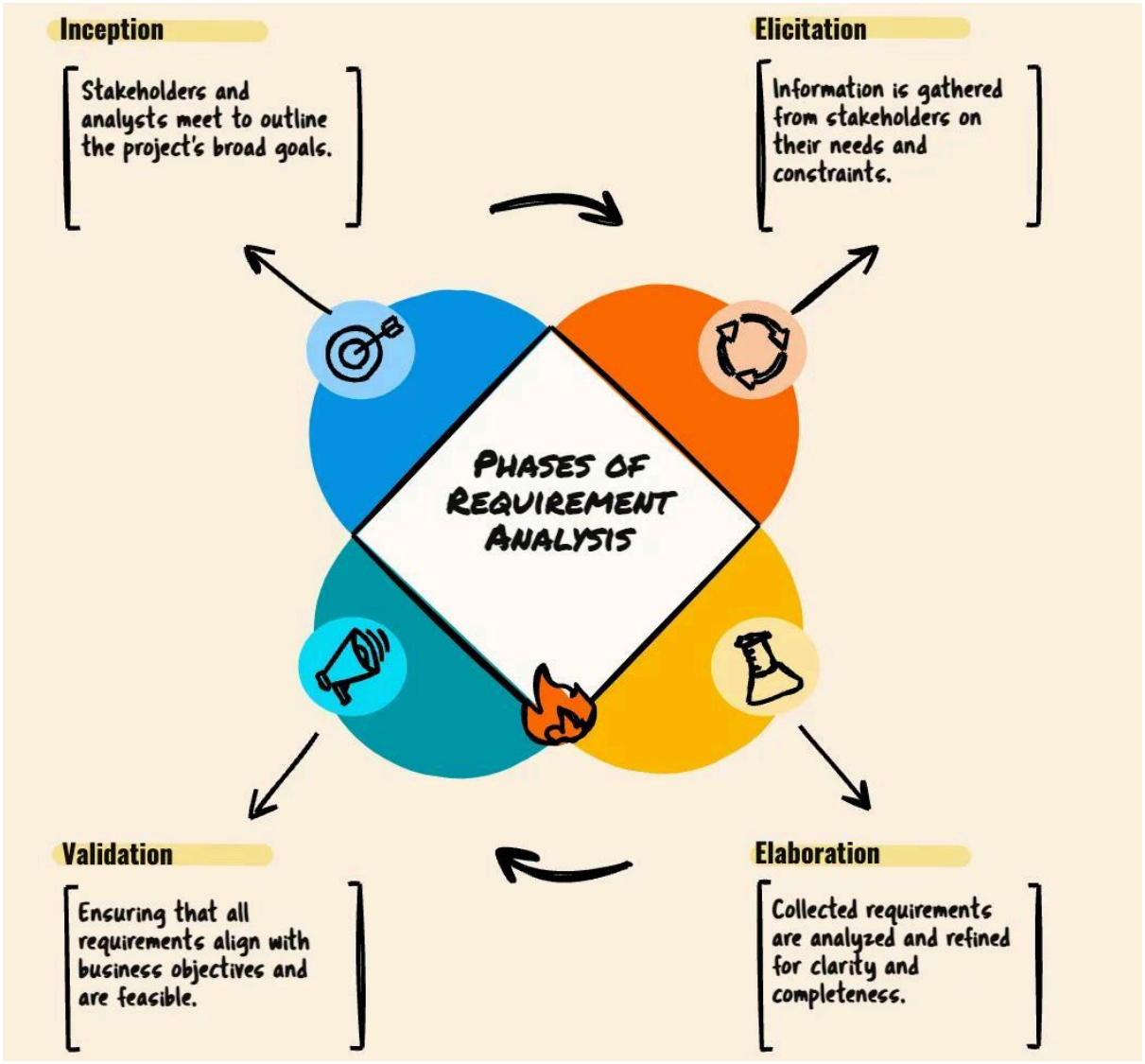
1. Requirement Gathering
   ↓
2. Feasibility Study
   ↓
3. System Design
   ↓
4. Development (Coding)
   ↓
5. Testing
   ↓
6. Deployment
   ↓
7. Maintenance

Phase 1: Requirement Gathering & Analysis

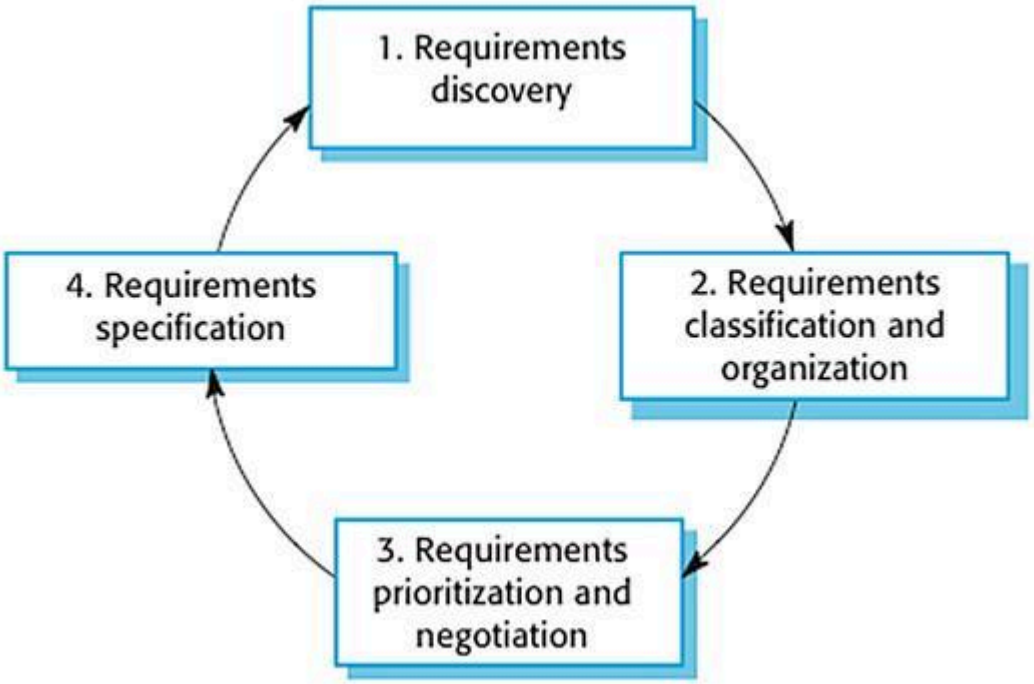

Client Requirement Analysis IT Project




Identify
Stakeholders


Inception
Stakeholders and analysts meet to outline the project's broad goals.
Elicitation
Information is gathered from stakeholders on their needs and constraints.
PHASES OF REQUIREMENT ANALYSIS
Validation
Ensuring that all requirements align with business objectives and are feasible.
Elaboration
Collected requirements are analyzed and refined for clarity and completeness.


BUSINESS REQUIREMENTS SPECIFICATION
Strategic Goals
Business Needs
User Stories
High-Level Workflows
Stakeholder Inputs
FUNCTIONAL SPECIFICATION
Functional Modules
Data Flows
UI Mockups
Test Cases
Logic/Rules
Test Cases



Definition

Requirements are collected from customers and stakeholders.

Objective

Understand what the customer needs.

Participants

- Client
- Business Analyst (BA)
- Product Owner
- Project Manager
- Stakeholders

Activities

- Client meetings
- Requirement gathering
- Documentation
- Requirement analysis

Deliverable

SRS (Software Requirement Specification)

Example

Customer says:

> "I need an online shopping application with payment gateway and order tracking."

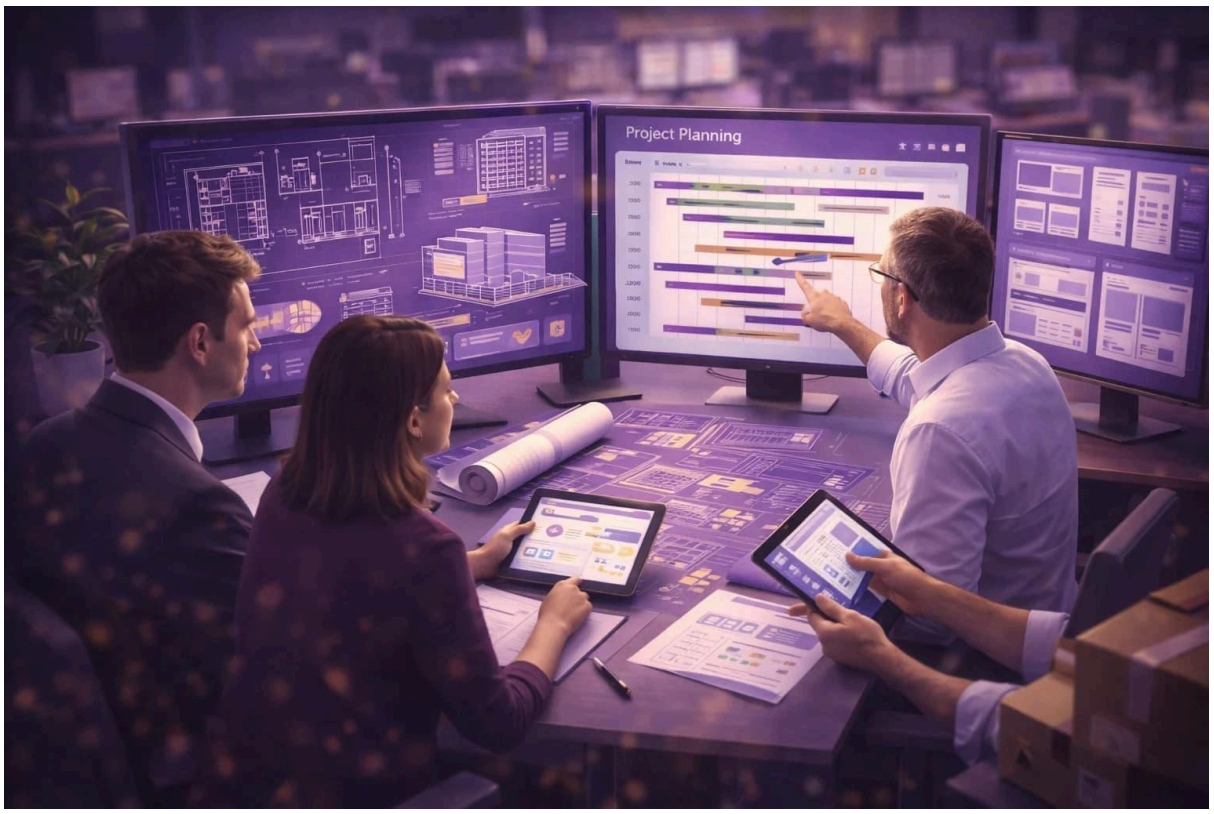
Phase 2: Feasibility Study





TECHNICAL FEASIBILITY

- Technical Feasibility assessment is focused on the present technical resource available in the organization.
- It studies if the technical resources including the technical team are capable of converting the ideas into working system.
- It also evaluates the hardware and the software requirement of the proposed system.





Definition

Determine whether the project is possible and worth developing.

Objective

Check cost, technology, resources, and timeline.

Participants

- Project Manager
- Business Analyst
- Technical Architect
- Finance Team

Activities

Technical Feasibility

Can technology support the project?

Economic Feasibility

Is budget available?

Operational Feasibility

Can users operate the system?

Schedule Feasibility

Can it be completed on time?

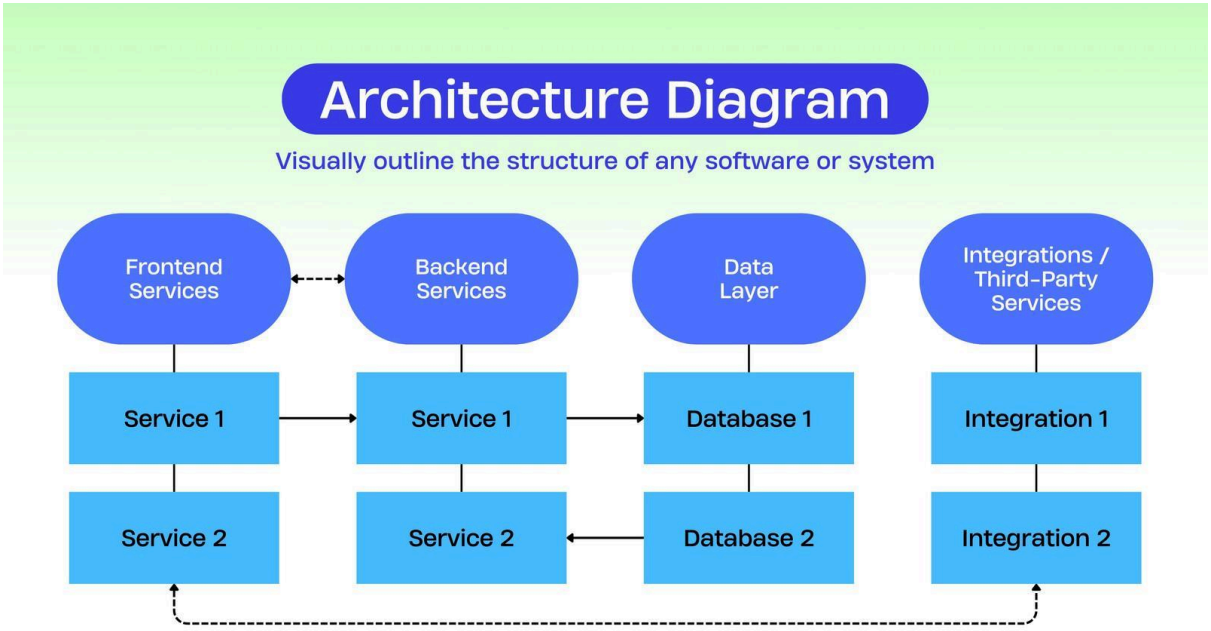
Deliverable

Feasibility Report

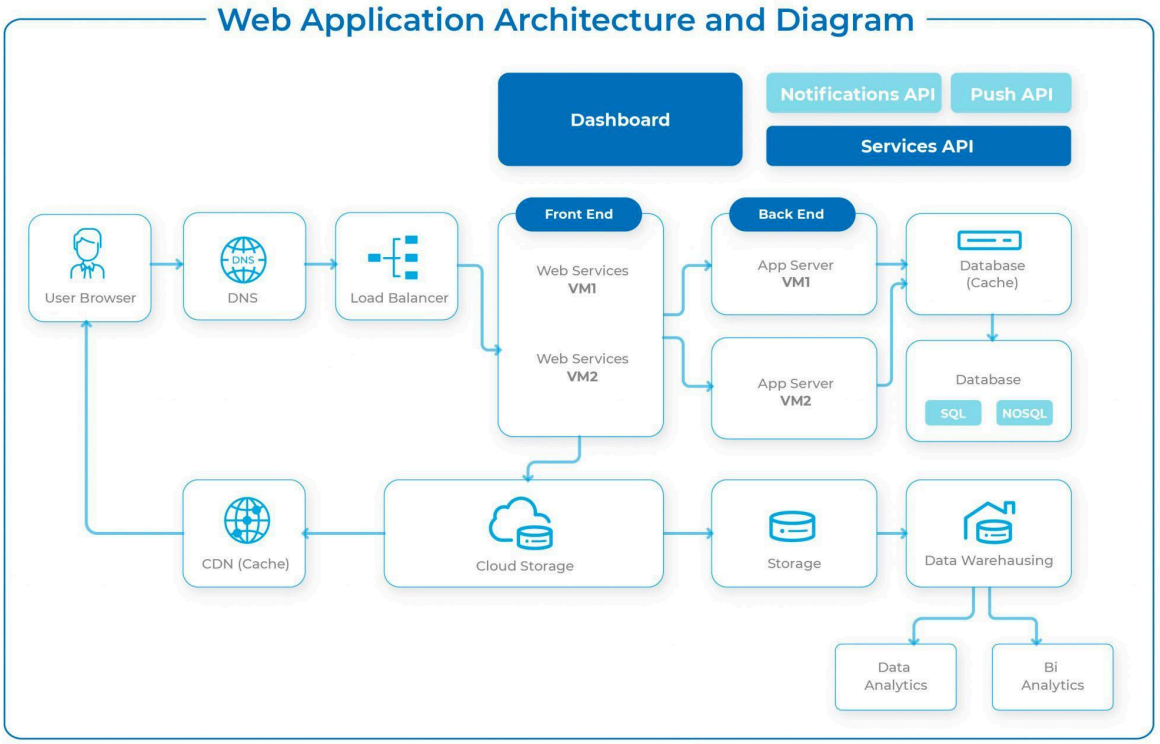
Example

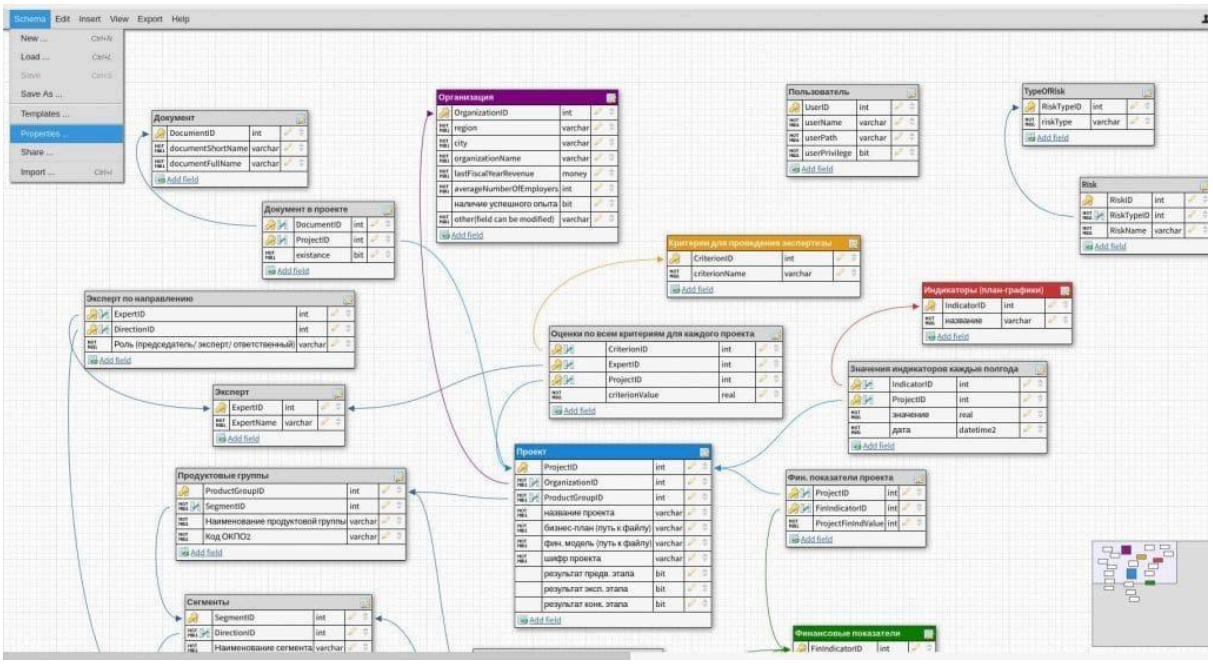
Building an AI chatbot may require cloud servers and AI models. Team checks budget and technical capability.

Phase 3: System Design


Architecture Diagram
Visually outline the structure of any software or system
Frontend Services
Backend Services
Data Layer
Integrations / Third-Party Services
Service 1
Service 1
Database 1
Integration 1
Service 2
Service 2
Database 2
Integration 2

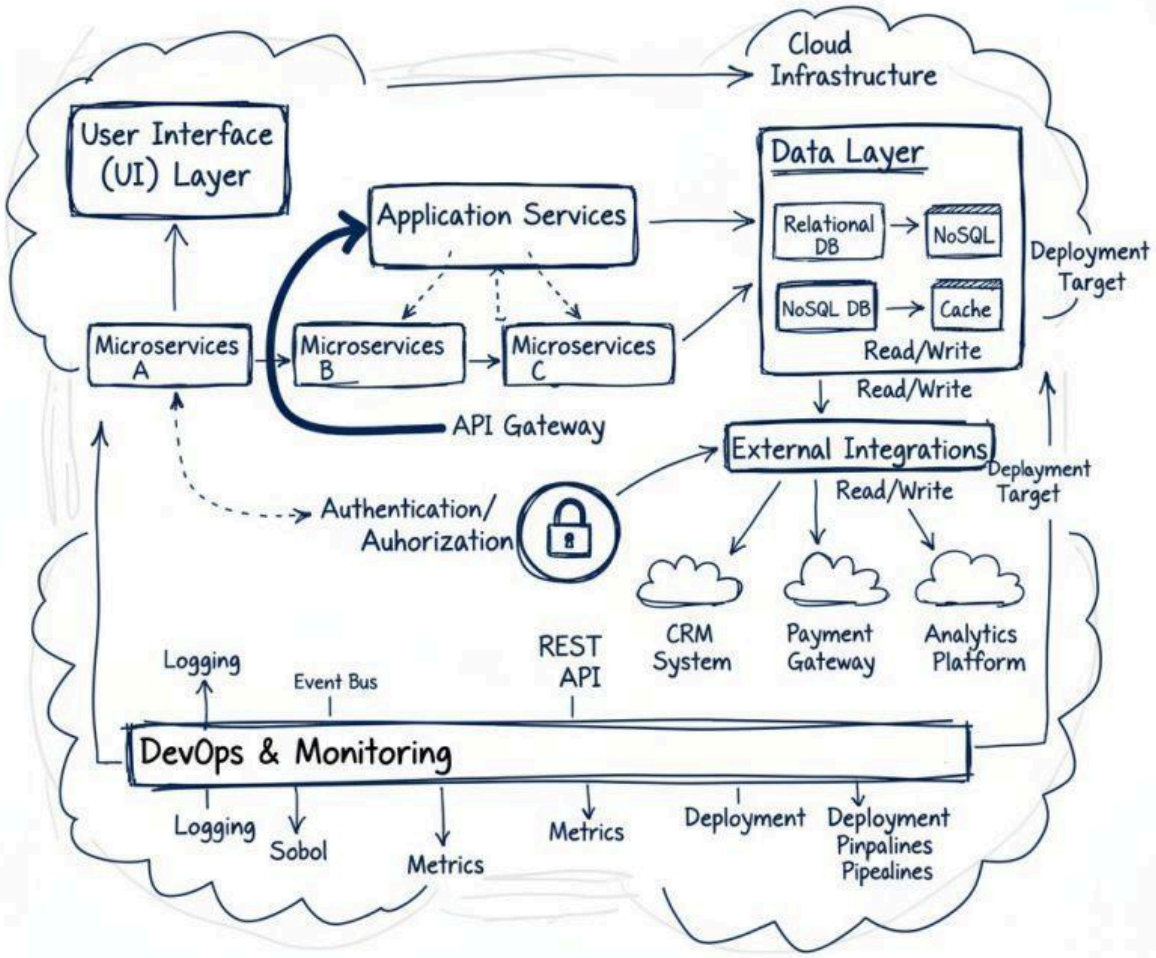

SYSTEM DESIGN

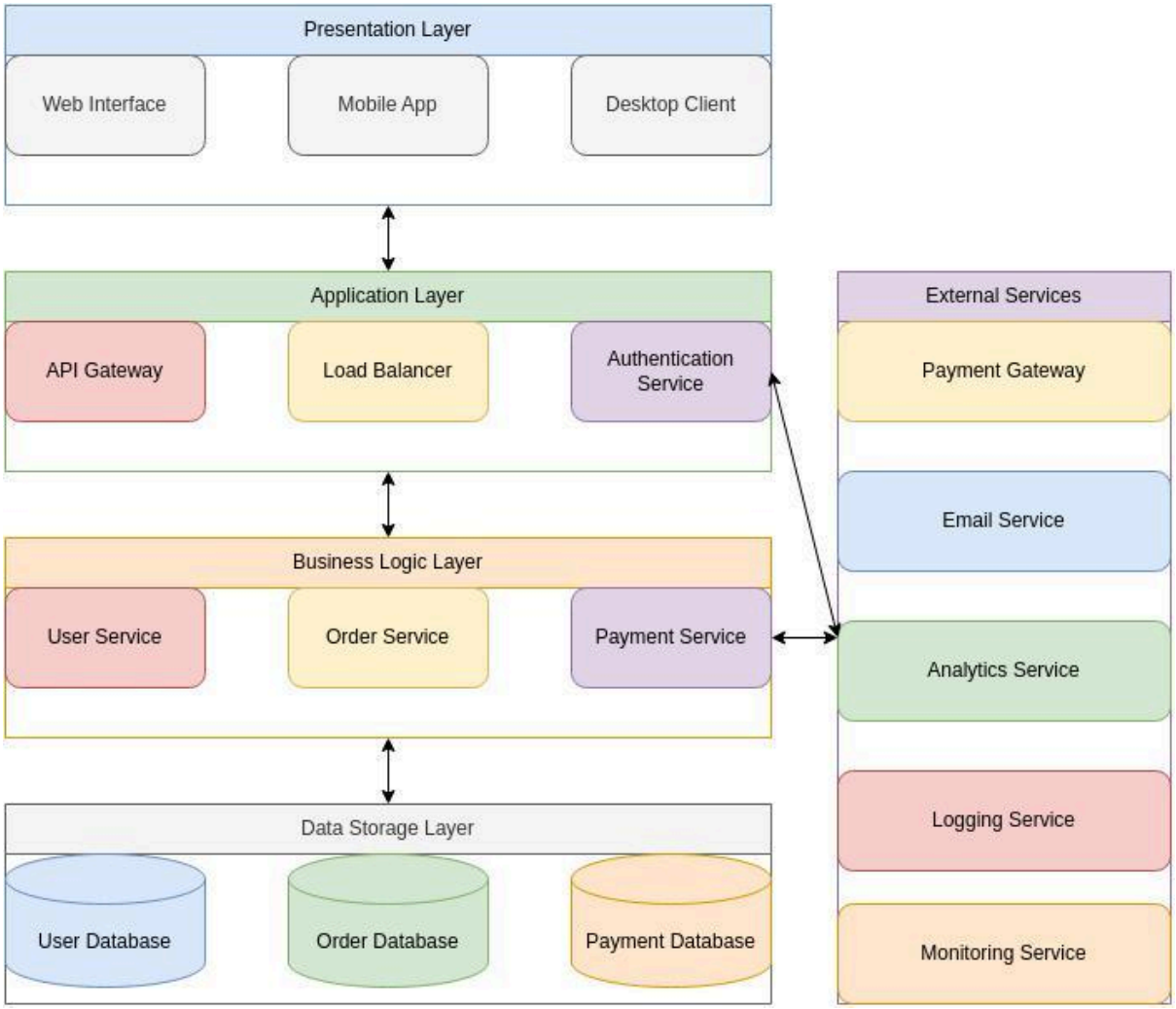

Web Application Architecture and Diagram
Dashboard
Notifications API
Push API
Services API
Front End
Back End
User Browser
DNS
Load Balancer
Web Services VM1
Web Services VM2
App Server VM1
App Server VM2
Database (Cache)
Database
SQL
NOSQL
CDN (Cache)
Cloud Storage
Storage
Data Warehausing
Data Analytics
Bi Analytics

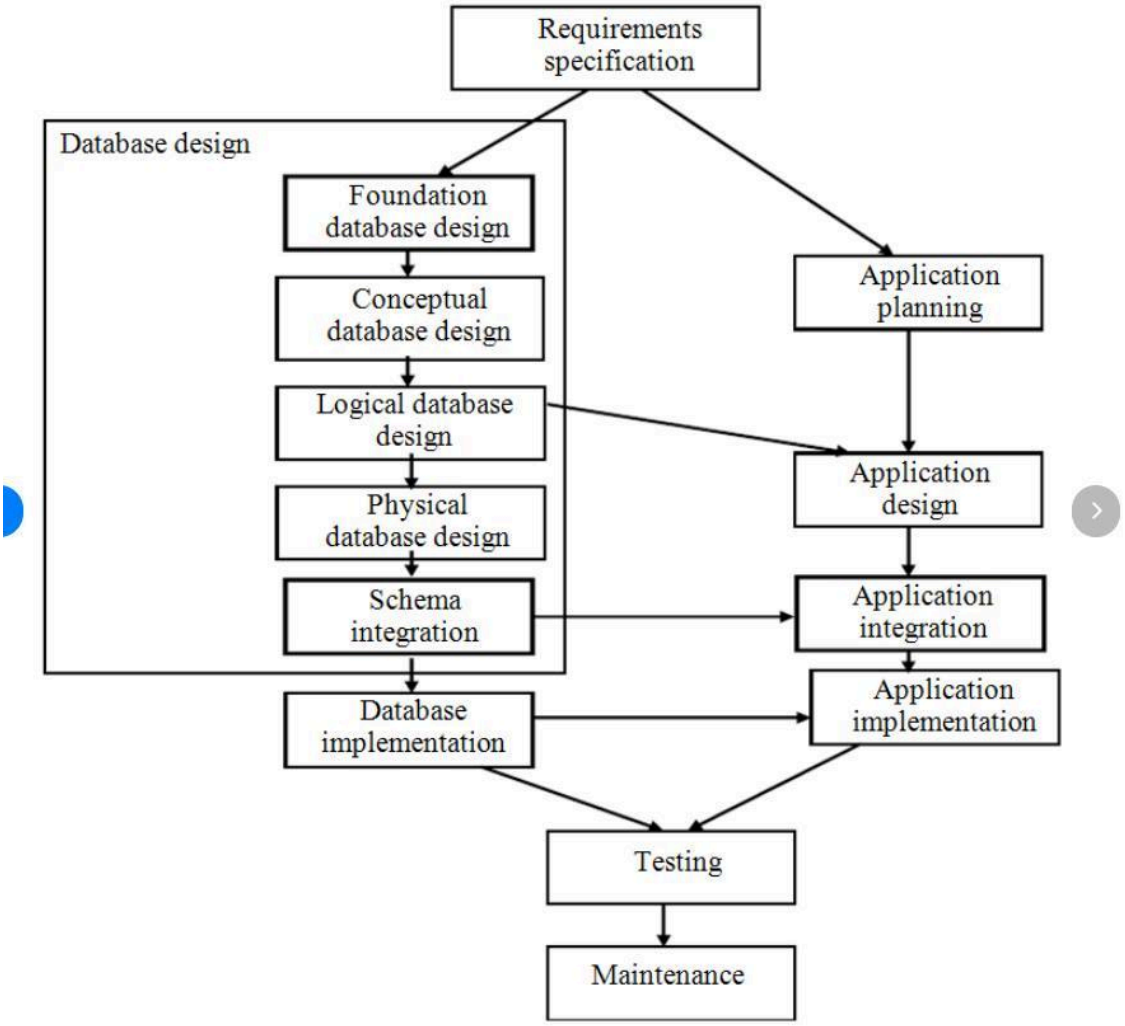



SYSTEM ARCHITECTURE :
USER
API
DATABASE
RESULT

Solution Architecture Blueprint - Initial Draft




Presentation Layer
Web Interface
Mobile App
Desktop Client
Application Layer
API Gateway
Load Balancer
Authentication Service
Business Logic Layer
User Service
Order Service
Payment Service
Data Storage Layer
User Database
Order Database
Payment Database
External Services
Payment Gateway
Email Service
Analytics Service
Logging Service
Monitoring Service



Definition

Convert requirements into technical design.

Objective

Create the software blueprint.

Participants

- Solution Architect
- Technical Architect
- UI/UX Designer
- Database Designer
- Senior Developers

Activities

- Architecture design
- Database design
- UI design
- API design

Deliverables

- HLD (High Level Design)
- LLD (Low Level Design)

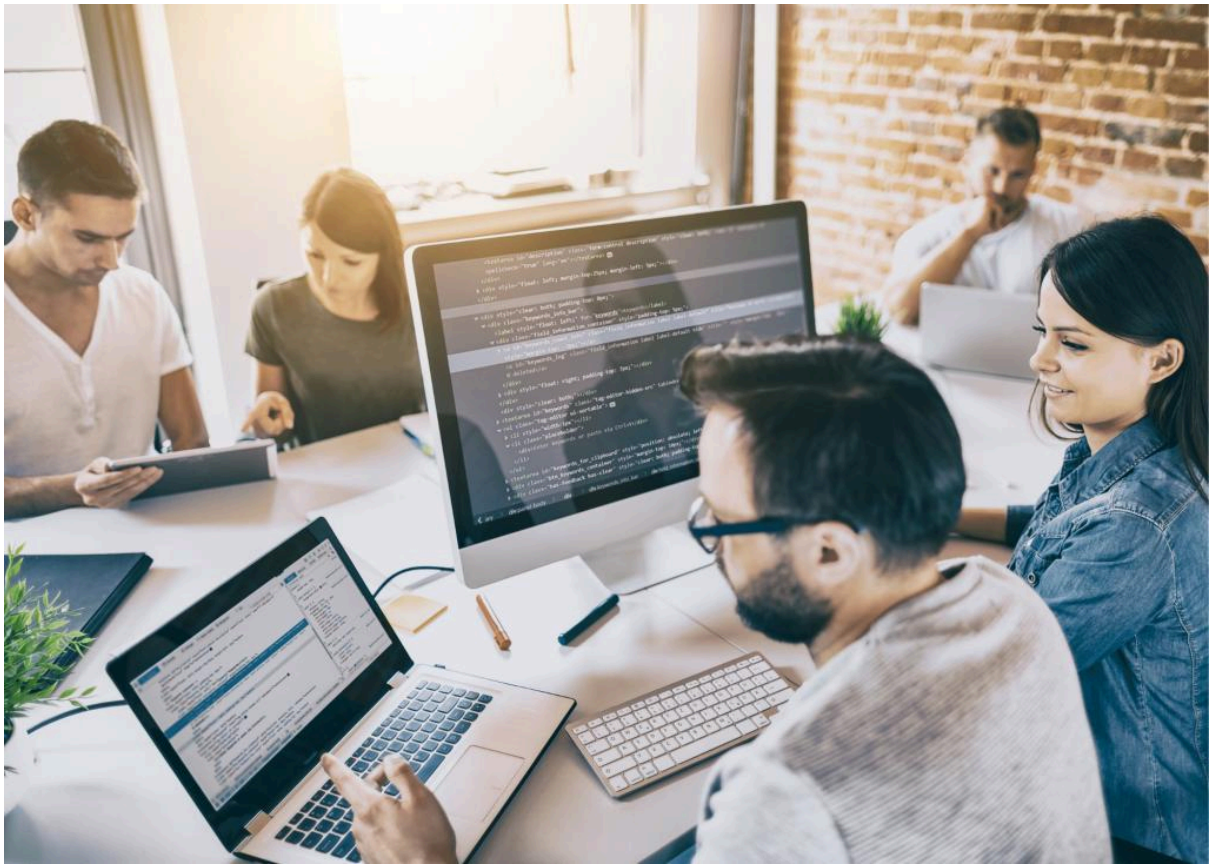
Example

E-commerce application design:

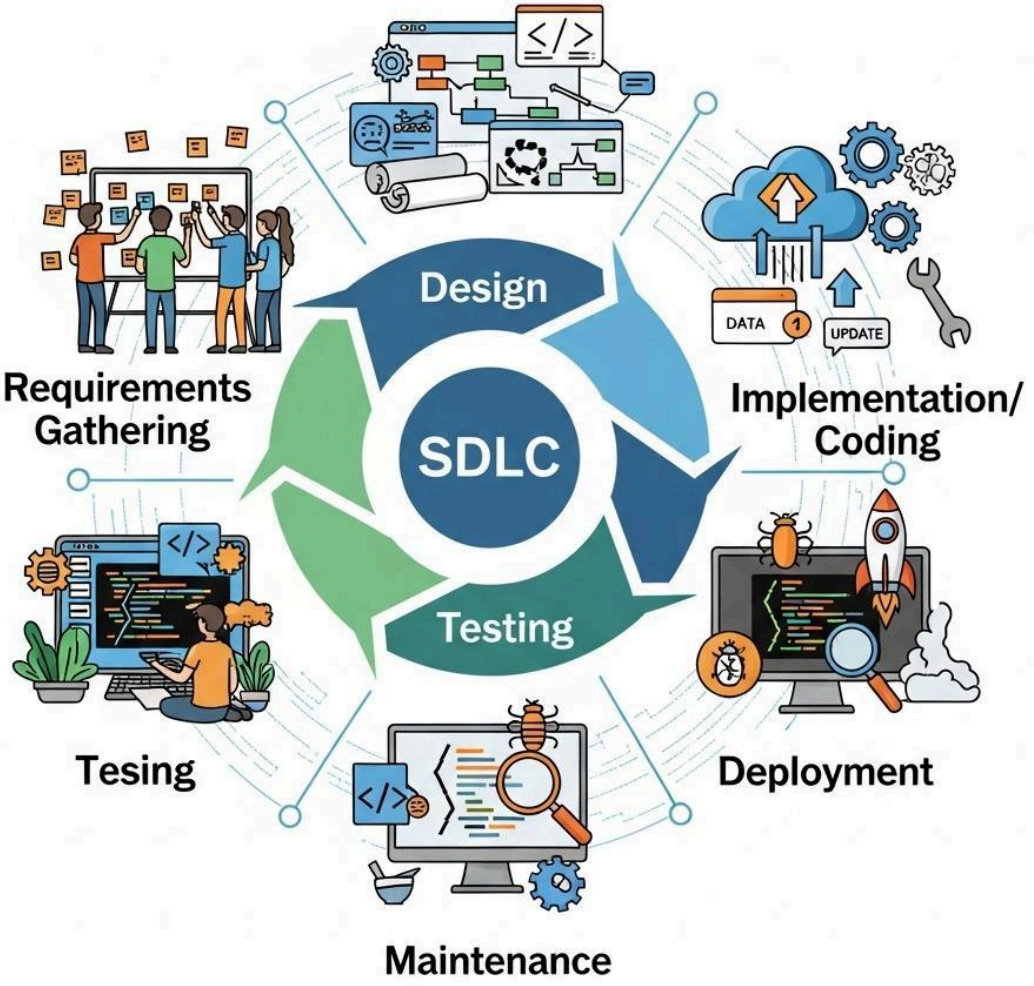
- Login Module
- Product Module
- Cart Module
- Payment Module

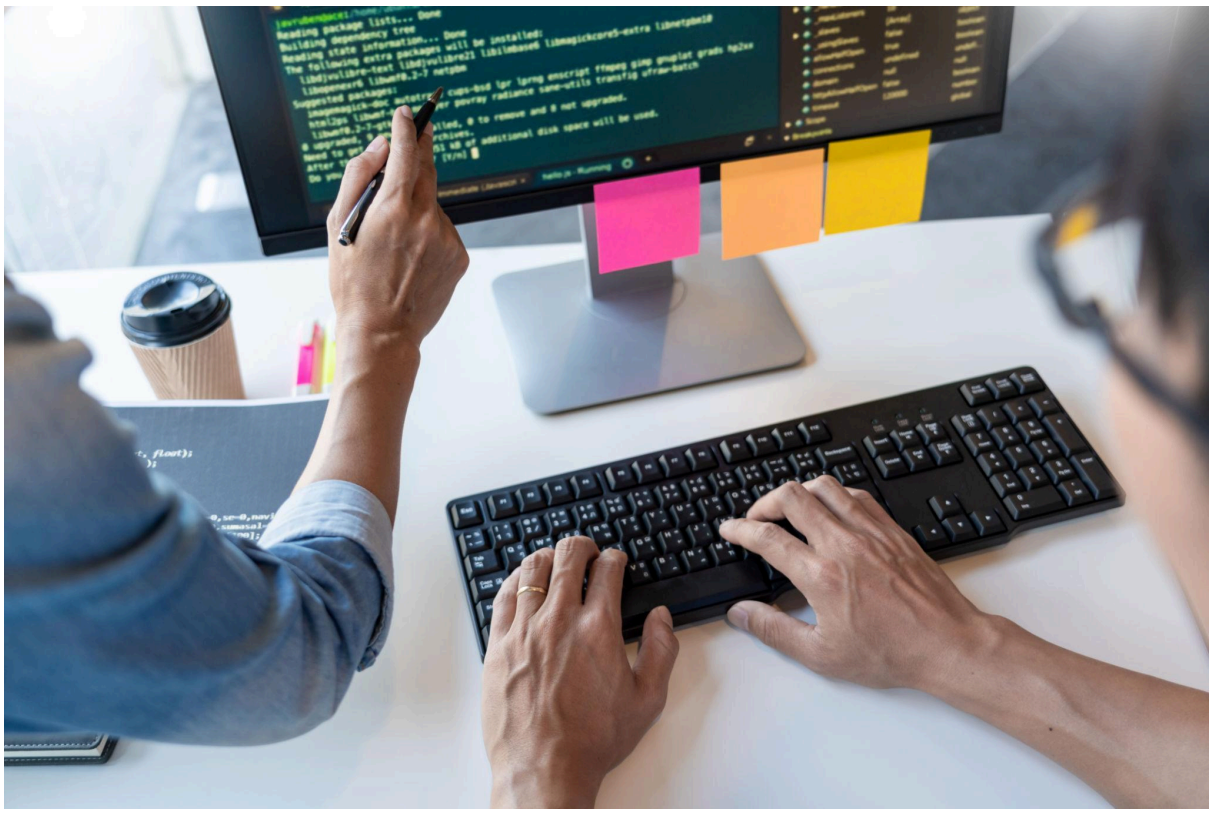
Phase 4: Development (Coding)





Software Development Lifecycle







Definition

Actual software coding is done.

Objective

Convert design into working software.

Participants

- Developers
- Technical Leads
- DevOps Engineers

Activities

- Coding
- Unit Testing
- Code Reviews
- Git Commit

Deliverable

Working Application

Example

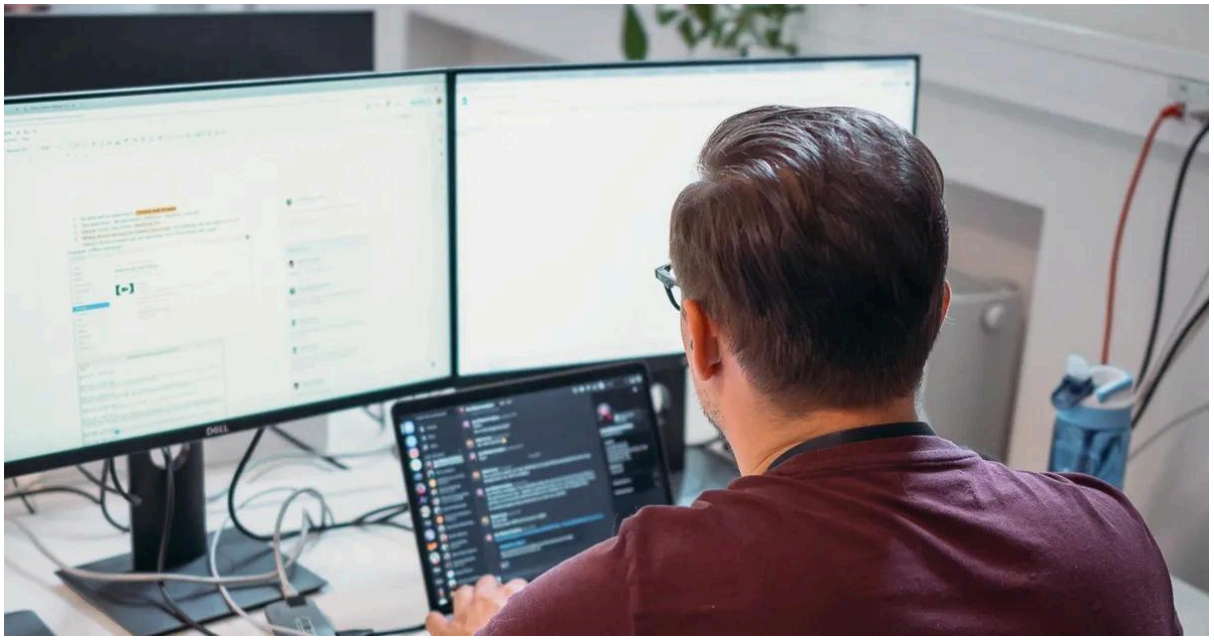
Developer creates:

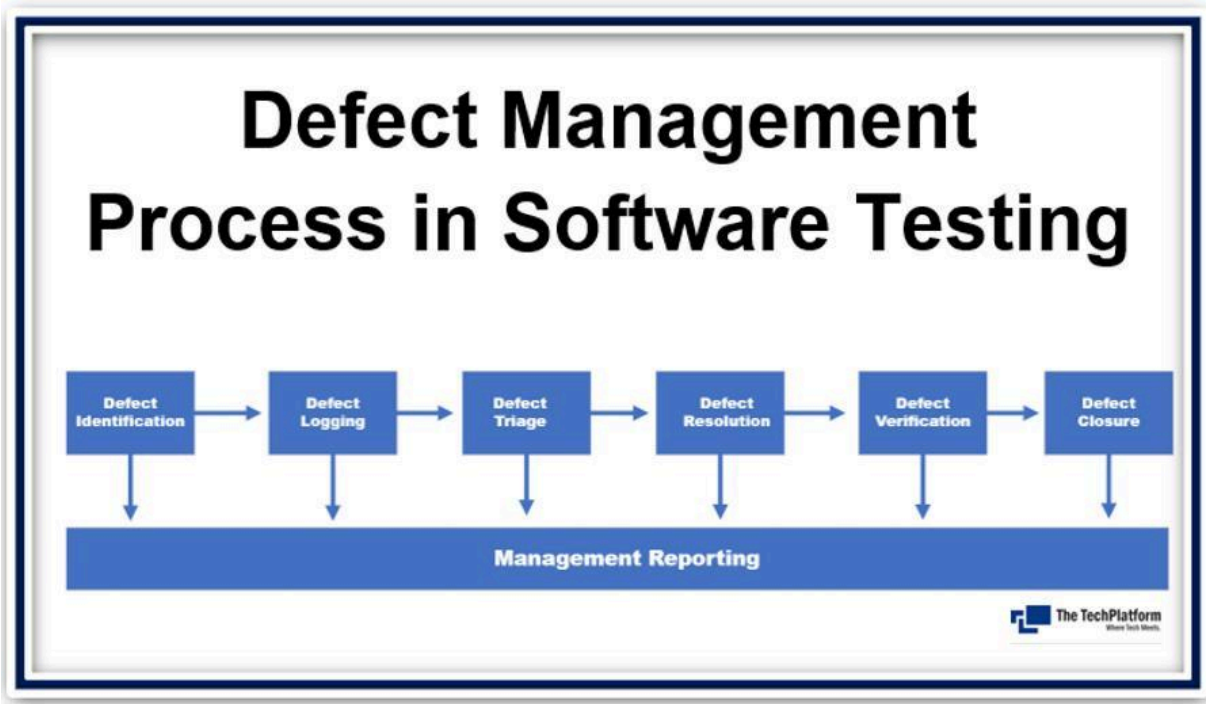
- Login Page
- Product Search
- Shopping Cart
- Payment Integration

Phase 5: Testing

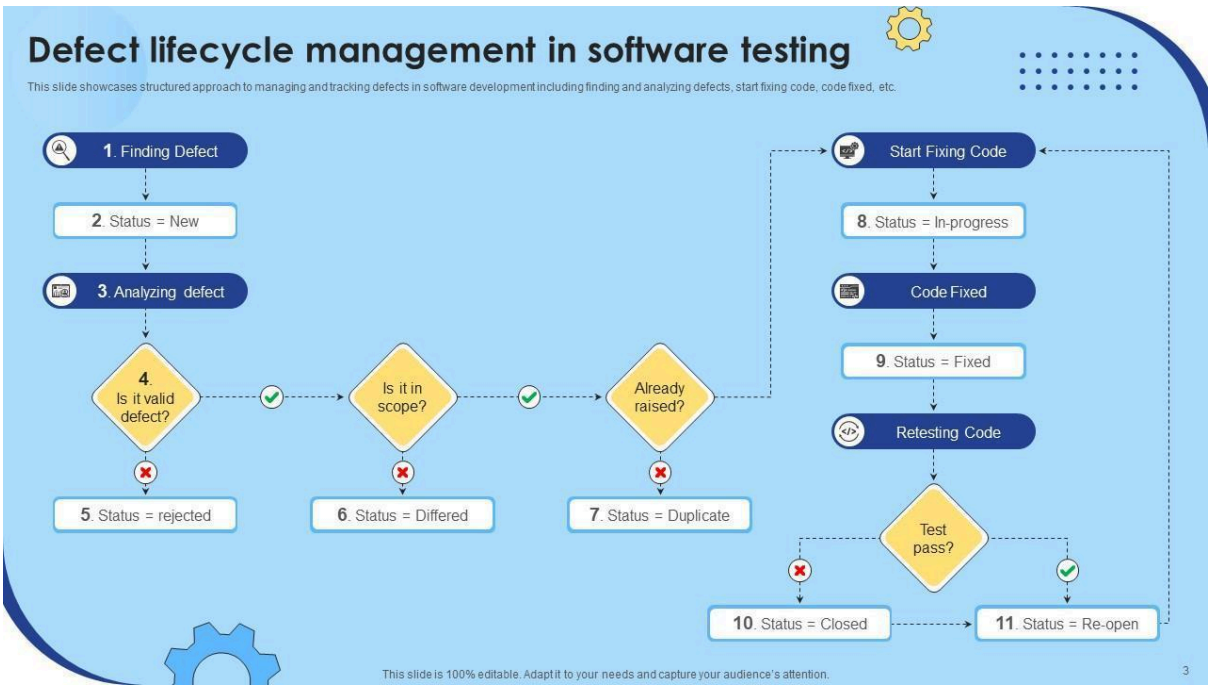

QA


Login Test
Check Data
ERROR




Defect Management Process in Software Testing
Defect Identification
Defect Logging
Defect Triage
Defect Resolution
Defect Verification
Defect Closure
Management Reporting
The TechPlatform


95%
PASSED



Definition

Verify that software works correctly and meets requirements.

Objective

Find and fix defects.

Participants

- QA Engineers
- Test Engineers
- Automation Testers
- UAT Users

Activities

Functional Testing

Checks functionality.

Integration Testing

Checks module interaction.

System Testing

Tests complete application.

Regression Testing

Ensures old features still work.

UAT Testing

Customer validation.

Deliverables

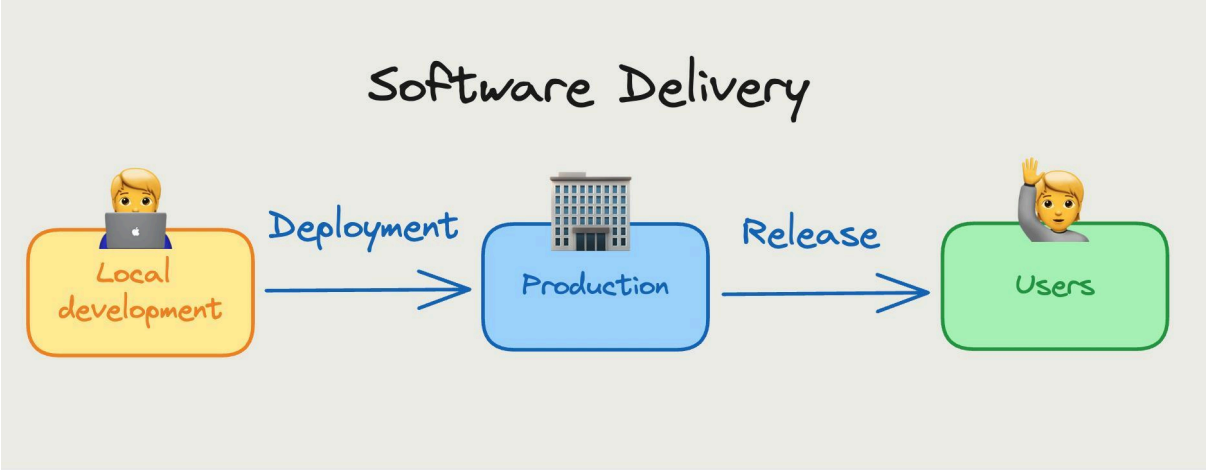
- Test Cases
- Test Reports
- Defect Reports

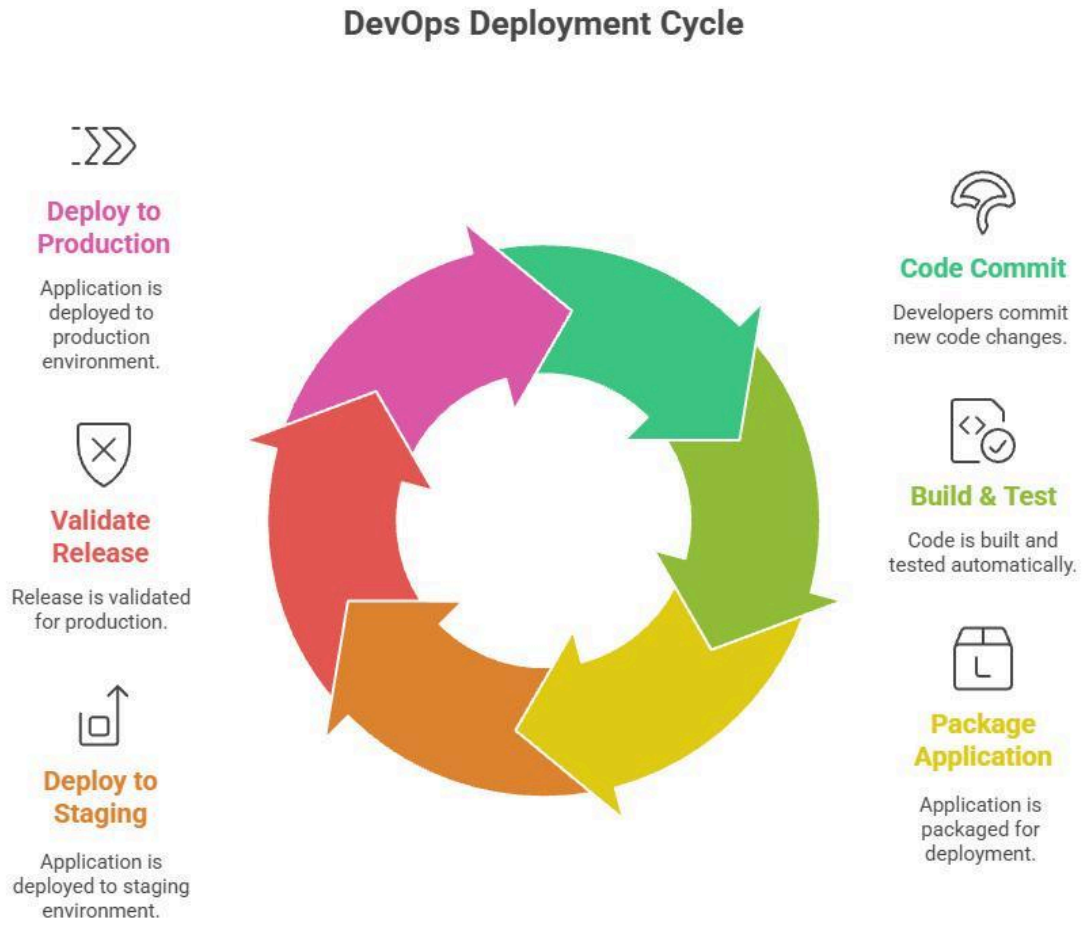
Example

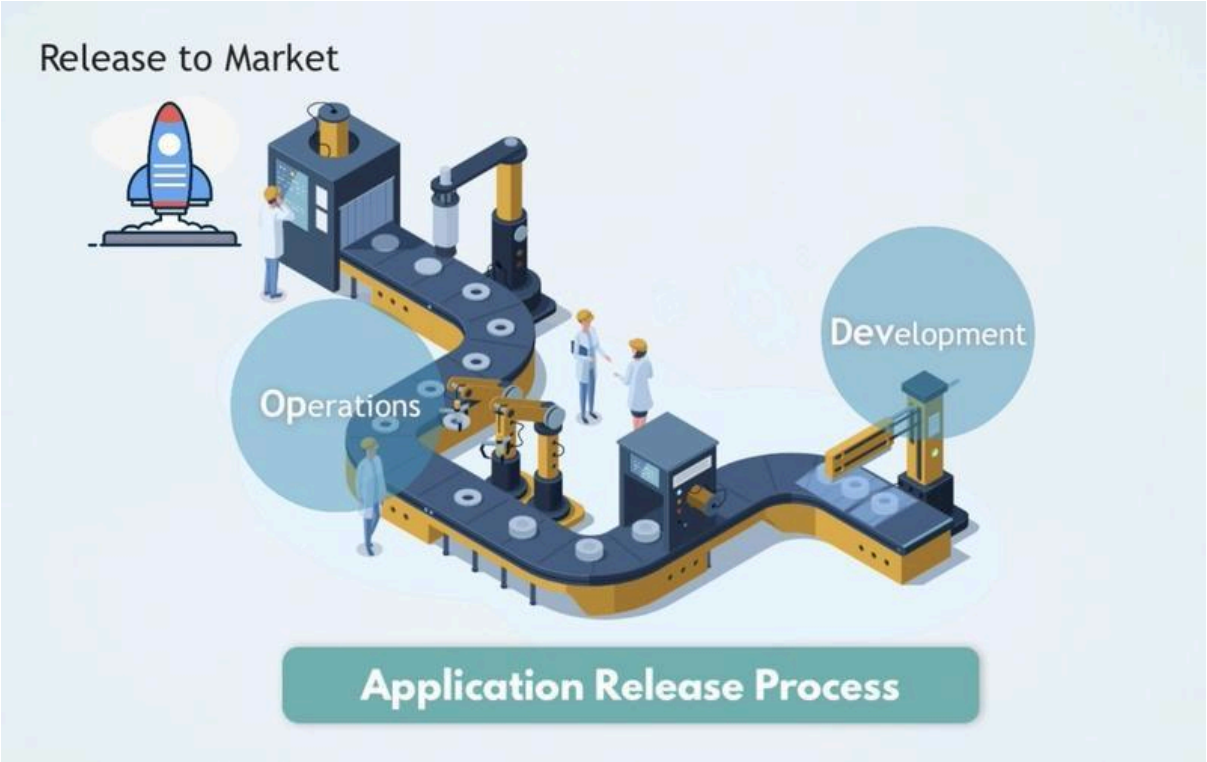
Tester verifies:

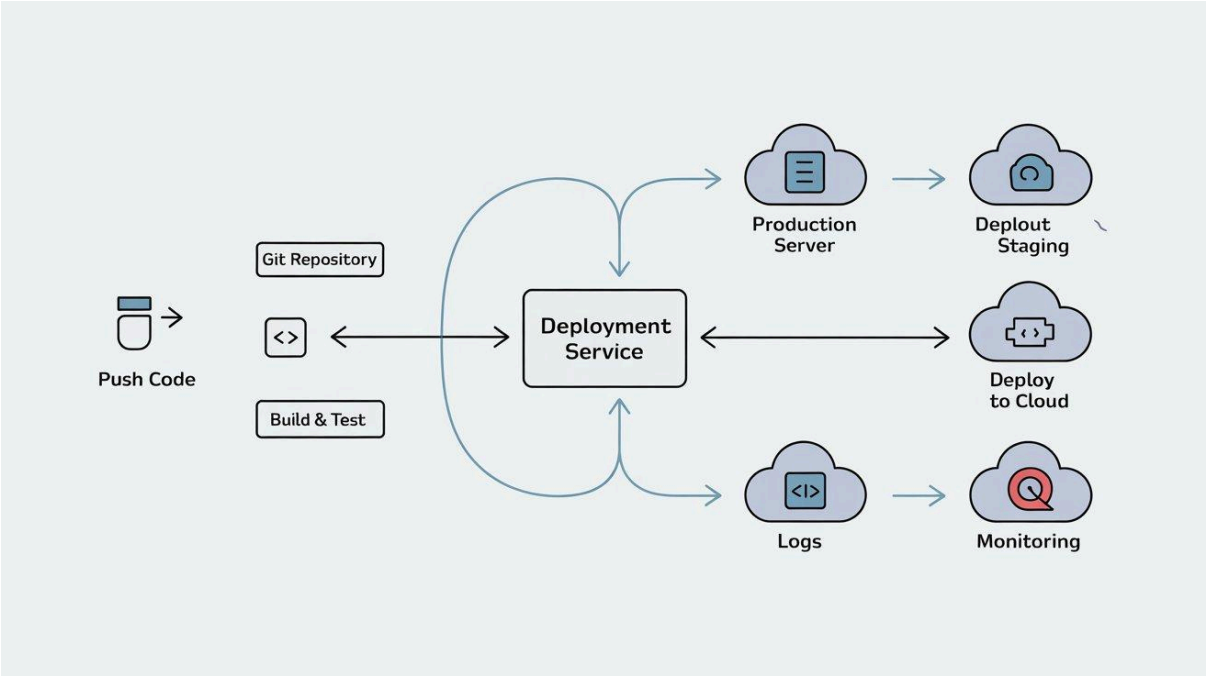
- Login works
- Payment succeeds
- Order tracking displays correctly

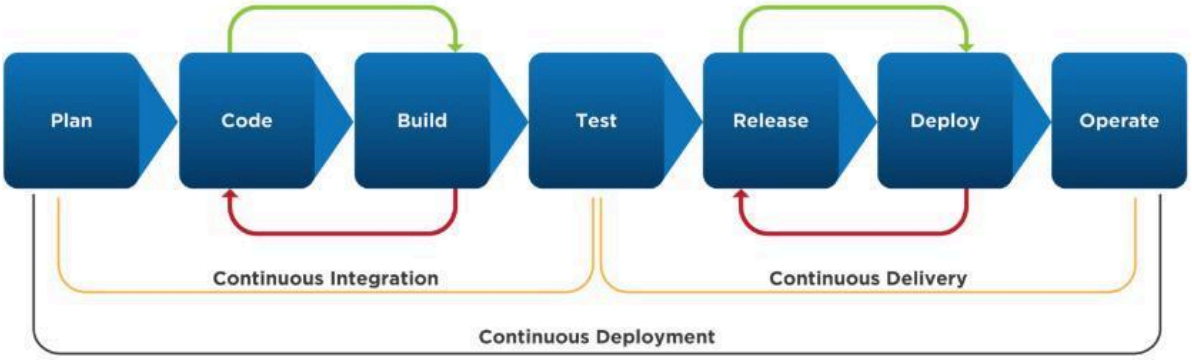
Phase 6: Deployment




DevOps Deployment Cycle
Deploy to Production
Application is deployed to production environment.
Validate Release
Release is validated for production.
Deploy to Staging
Application is deployed to staging environment.
Code Commit
Developers commit new code changes.
Build & Test
Code is built and tested automatically.
Package Application
Application is packaged for deployment.


Release to Market
Development
Operations
Application Release Process





Planning
- Requirement finalization
- Updates & new changes
- Architecture & design
- Task assignment
- Timeline finalization

Code
- Development
- Configuration finalization
- Check-in source code
- Static-code analysis
- Automated review & peer review

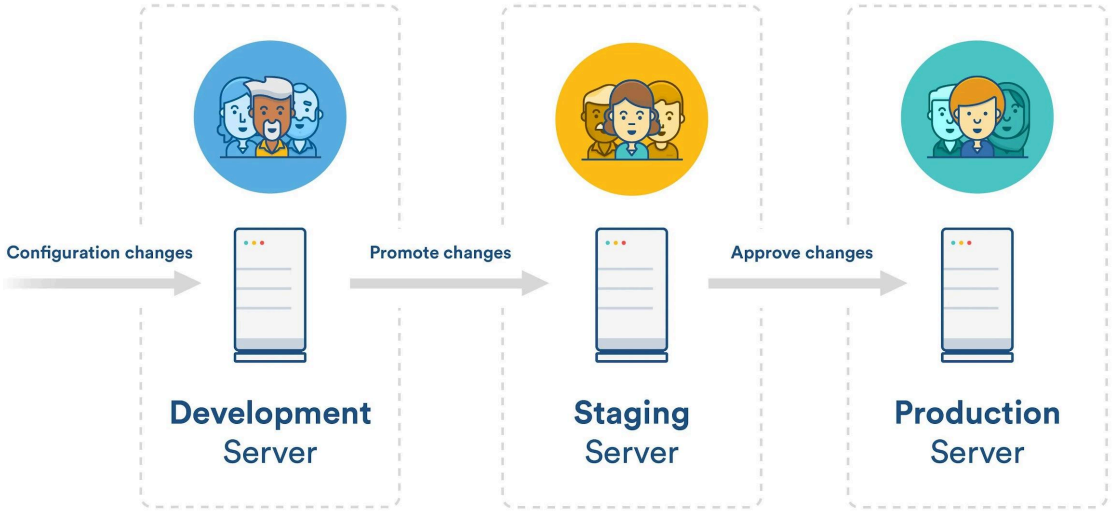
Build
- Compile code
- Unit testing
- Code-metrics
- Build container images or package
- Preparation or update in deployment templated
- Create or update monitor dashboards

Test
- Integration test with other component
- Load & stress test
- UI testing
- Penetration testing
- Requirement testing

Release
- Preparing release notes
- Version tagging
- Code freeze
- Feature freeze

Deploy
- Updating the infrastructure i.e staging, production
- Verification on deployment i.e smoke tests

Operate
- Monitor designed dashboard
- Alarm triggers
- Automatic critical events handler
- Monitor error logs





Definition

Application is released to production.

Objective

Make software available to end users.

Participants

- DevOps Team
- Release Manager
- System Administrator
- Support Team

Activities

- Production deployment
- Server configuration
- Database migration
- Smoke testing

Deliverable

Live Application

Example

Amazon launches a new shopping feature for customers.

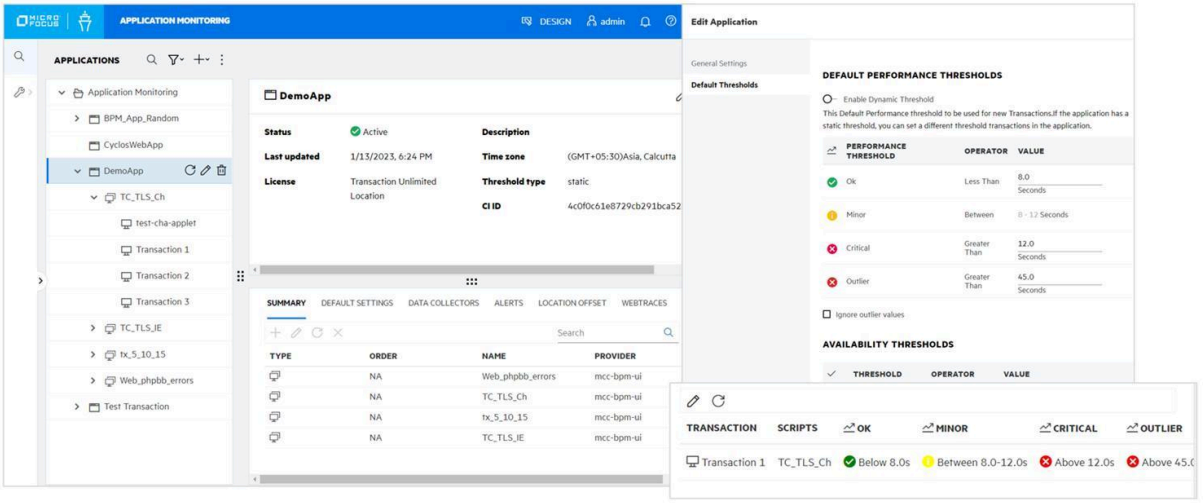
Phase 7: Maintenance











Definition

Support and improve software after release.

Objective

Keep software running smoothly.

Participants

- Support Team

- Developers
- DevOps Team
- Product Team

Activities

- Bug fixing
- Performance tuning
- Security updates
- Feature enhancements

Deliverable

Updated Versions

Example

Customer requests:

- Dark mode
- Faster search
- New payment methods

Team implements updates in future releases.

SDLC Summary Table

Phase	Purpose	Participants	Deliverable
Requirement Gathering	Collect requirements	BA, Client, PM	SRS
Feasibility Study	Check project viability	PM, Architect, BA	Feasibility Report
System Design	Create blueprint	Architect, Designer	HLD, LLD
Development	Write code	Developers	Application
Testing	Find defects	QA Team	Test Report
Deployment	Release software	DevOps Team	Live Application

Maintenance	Support and enhancements	Support Team	Updated Software

Interview Question

Q: What are the 7 phases of SDLC?

Answer:

1. Requirement Gathering & Analysis
2. Feasibility Study
3. System Design
4. Development (Coding)
5. Testing
6. Deployment
7. Maintenance

These phases help deliver high-quality software in a structured, cost-effective, and reliable manner.